

Sahil Sanghvi

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EDUCATION & AWARDS

University of Victoria, Faculty of Engineering and Computer Science

B.S. Computer Science, Software Systems Specialization

Victoria, BC

Sept. 2023 – Present

Runner-Up (Google Gemini Track) & Best .tech Domain — nwHacks 2026

January 2026

People's Choice Award, Victoria BC — NASA International SpaceApps Challenge 2025

October 2025

Assistant Guest Lecturer — Augmented Reality and C, DA-IICT [[Presentation Media](#)]

August 2019

TECHNICAL WORK EXPERIENCE

Protean eGov Technologies

Mumbai, India (Remote)

AI Intern — CIO Department

May 2026 – Present

- Designed the architecture for an agentic AI platform automating end-to-end social media marketing across **5 platforms**, technical SEO, and Generative Engine Optimization (GEO), eliminating an estimated **50–60+ hours** of monthly manual work.
- Architected a **6-agent** LangGraph orchestration system (Trend, Competitor, Performance, Prediction, Strategy, Calendar) with a custom LLM Gateway for per-agent model tiering between Claude and GPT-4o, hard cost caps, and hot-swappable routing.
- Backed the platform with a Redis caching layer and pgvector semantic search over **30 competitor handles** for RAG-powered competitive intelligence.
- Built the Performance agent to run automated technical SEO audits and generate RAG-grounded blog content, expanding discoverability for Digital Public Infrastructure products serving **9.4+ crore** subscribers.
- Shipped a production NLP/RAG chatbot, now in UAT and impacting **1,000+** employees, with role-based access control, hallucination guardrails, and human-in-the-loop escalation.
- Architected a serverless, event-driven GCP deployment (Cloud Run Jobs, Cloud Scheduler, Secret Manager) scaling to zero at **\$260–280/month** with Microsoft Entra ID SSO/OIDC and RBAC.
- Authored the Solutions & Architecture Document, Security Protocols, and NIST CSF sign-off checklist, bringing the platform into DPDP Act 2023 compliance.

Stealth Startup

Remote

Head Software Engineer

May 2026 – Present

- Leading a team of **3** engineers building **3** products end-to-end — a mentor-mentee marketplace, an AI interview practice bot, and a career-services platform.
- Owning system design for an Airbnb-style marketplace connecting mentees with mentors for skills-based, hourly-rate sessions, covering discovery, booking, and payments.
- Built an AI interview practice bot on FastAPI with server-rendered Jinja2 templates, using browser-native `getUserMedia/MediaRecorder` for live video/audio capture and the Web Speech API for real-time voice synthesis.
- Designed multi-provider LLM and speech-to-text failover chains (Groq Llama 3.3 70B, Gemini 2.0 Flash, GPT-4o-mini; Groq Whisper large-v3-turbo) for resilient interview scoring and transcription.
- Integrated Razorpay HMAC-verified payments to gate access to interview results.
- Built the startup's WordPress marketing site and a custom PHP admin portal integrated with the interview bot via a REST bridge, letting recruiters manage interviews, candidates, and payments from one dashboard.

RBC Borealis — Let's Solve It Mentorship

Remote, Canada

AI & Machine Learning Research Fellow

March 2026 – Present

- Selected for RBC Borealis' competitive Spring 2026 cohort; developed a real-world AI/ML solution in a team of **4 mentees** alongside **2 senior RBC AI researchers**.
- Surfaced disengagement signals across **20+** behavioural features via exploratory data analysis on **1,500+** longitudinal youth health engagement records spanning **3+ years**.
- Isolated **4+** distinct user cohorts by reducing high-dimensional engagement data to **2D embeddings** with UMAP clustering in scikit-learn.
- Engineered a PyTorch pipeline to flag at-risk youth navigating chronic illness, enabling earlier clinical intervention.
- Corrected inflated raw DAU from **40 to 27** for stakeholder and funder reporting by building a GA4 analytics pipeline with bot-aware filtering across **1,048 days** of data.
- Designed a multi-page Streamlit dashboard backed by **5 ML models** spanning forecasting, anomaly detection, and clustering, serving **4+** clinical and program teams.

HACKATHONS & COMPETITIONS

PrecheckAI — Runner-Up (Google Gemini Track) & Best .tech Domain, nwHacks 2026 | *React, TypeScript, FastAPI, scikit-learn*

- Shipped a healthtech MVP among **100+** competing teams in under **24 hours**, leading a **4-person** team building an AI-powered health insurance claims assistant.
- Architected a **4-stage** ML validation pipeline: Gemini Vision OCR, RAG-based policy retrieval, HistGradientBoosting risk scoring (**0–100** denial probability), and a remediation dashboard with citation-level fixes.
- Covered the full claim lifecycle across **3** user flows (Prevention, Pre-Submission, Appeal), validating submissions against **10+** required fields via FastAPI.

ExoDetect — People’s Choice Award, NASA International SpaceApps Challenge 2025 | *TypeScript, Next.js, FastAPI, Docker*

- Achieved **97.3%** accuracy classifying exoplanets from NASA light-curve data with ensemble ML models, outperforming baselines by **40%**.
- Cut API latency under **180 ms** and preprocessing compute time by **72%** by architecting a full-stack pipeline (FastAPI + Docker + Next.js) with vectorization and caching.

INDEPENDENT PROJECTS

HRMS Platform | *.NET 8, PostgreSQL, JWT, Docker, GitHub Actions, CI/CD* 2026

- Designing a multi-tenant HRMS platform on a .NET 8 backend for SMBs priced out of enterprise HR software.
- Building **7** frontend modules with a custom design system, backed by PostgreSQL with versioned migrations for multi-tenant data isolation.
- Integrating **4** core and **4** advanced AI features into recruiter-company matching and applicant workflows.
- Implementing JWT authentication, CORS policies, and rate limiting; set up CI/CD via Docker and GitHub Actions for deployment to Render/Railway or self-hosted environments.

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, TypeScript, Go, C, C++, SQL

Frontend & Web: React.js, Next.js, Tailwind CSS, HTML5, CSS3, Recharts, Chart.js, Responsive Design

Backend & APIs: FastAPI, RESTful APIs, Pydantic, SQLAlchemy, Alembic, Node.js, .NET 8, Microservices

AI/ML & Data: LangGraph, RAG, Embedding Models, Vector Search, LLM APIs (Claude, OpenAI GPT-4o, Gemini, Groq), scikit-learn, XGBoost, LightGBM, PyTorch, NumPy, Pandas, pgvector

Databases & Caching: PostgreSQL, SQL Server, MySQL, Redis, Tableau, Power BI

Cloud & DevOps: GCP (Cloud Run, Cloud Scheduler, Secret Manager, Artifact Registry), AWS, Azure, Docker, Kubernetes, CI/CD, Git/GitHub, Unix/Linux

Testing: JUnit, pytest, Selenium IDE, TDD, Mutation Testing, Property-Based Testing

AI-Assisted Dev: Cursor, Claude Code, GitHub Copilot, ChatGPT

VOLUNTEER ACTIVITIES AND COMMUNITY INVOLVEMENT

UVic UMANG ISA — Vice President

Victoria, BC

September 2023 – Present

- Led cross-cultural programming initiatives for **100+** students, preparing formal proposals for university departments and external stakeholders while coordinating events and managing budgets.
- Organized large-scale orientation events by managing timelines, coordinating logistics, and maintaining sponsorship and volunteer records using Google Workspace.